CNENGR115
Computers and Control
An Introduction to Feedback

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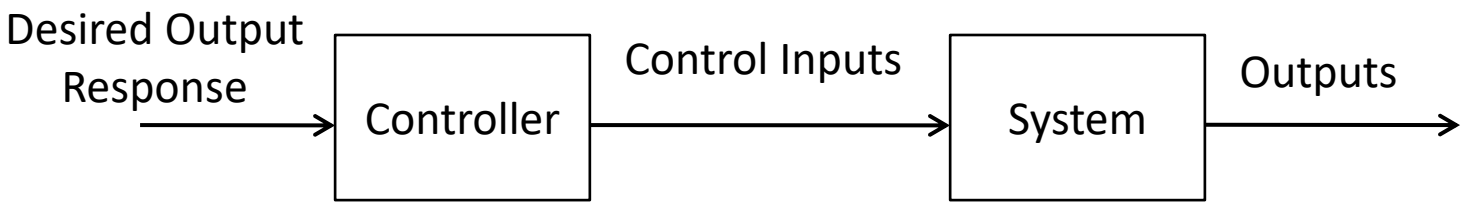
Control Lectures

Part 1 – Introduction

Part 2 – Mathematical Models

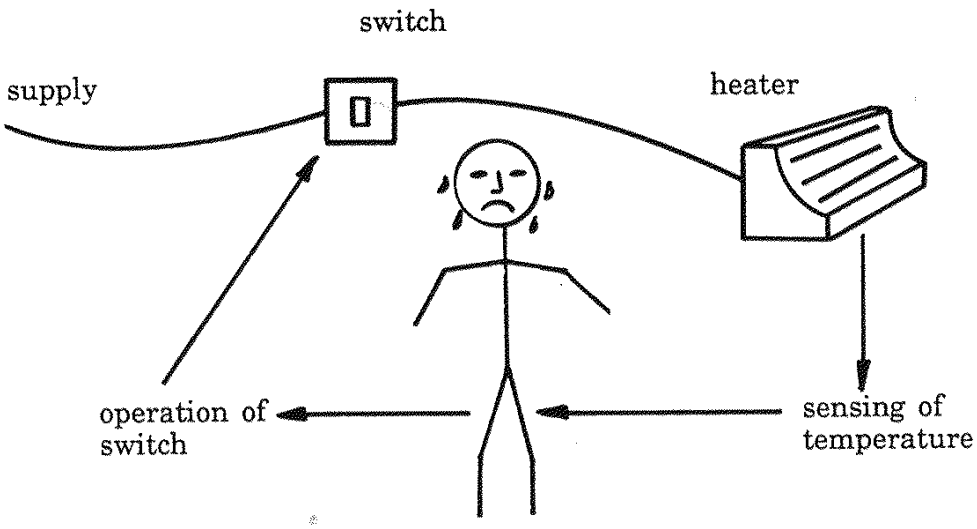
Part 3 – Control Actions

Open–Loop Control Systems

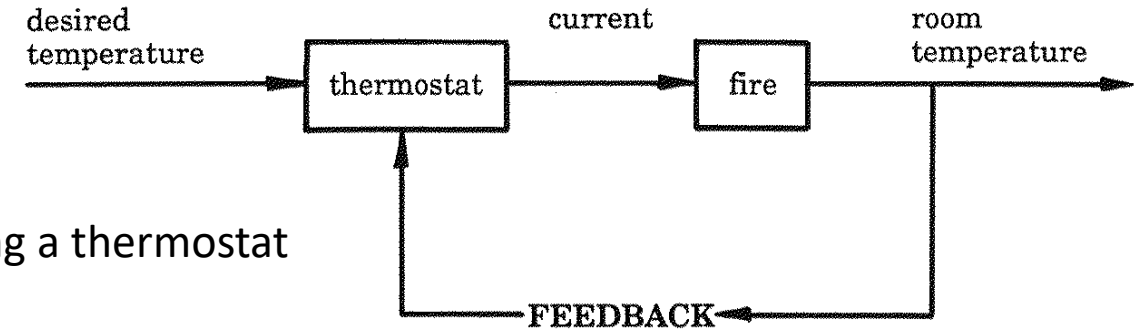


- Open–loop systems do **not** have feedback
- Example – an electric fire in a room
- Desired output response is a comfortable room temperature
- Controller is a human who turns heater on
- Without further human intervention, the room temperature (output) will eventually arrive at some constant level
- Room might still be too hot or too cold!

Closed–Loop Control Systems

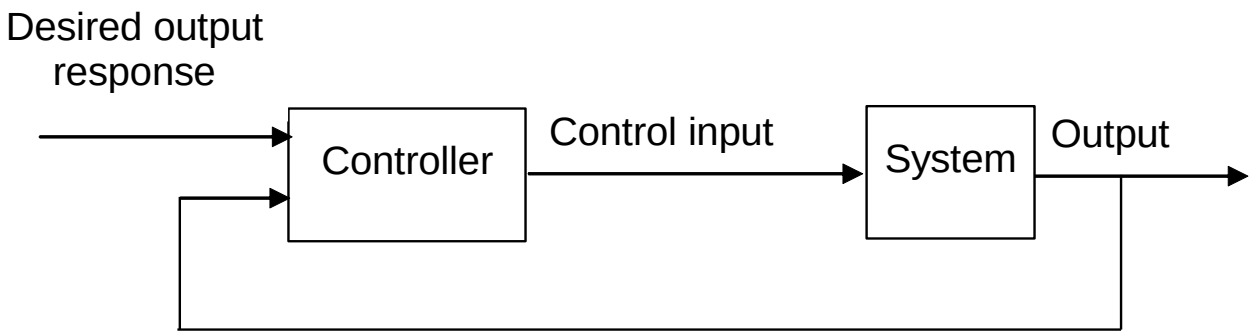


Feedback by human intervention



Automatic feedback using a thermostat

Closed–Loop Control Systems

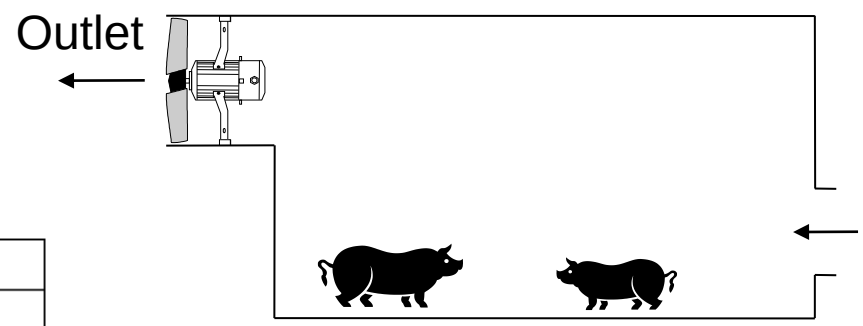


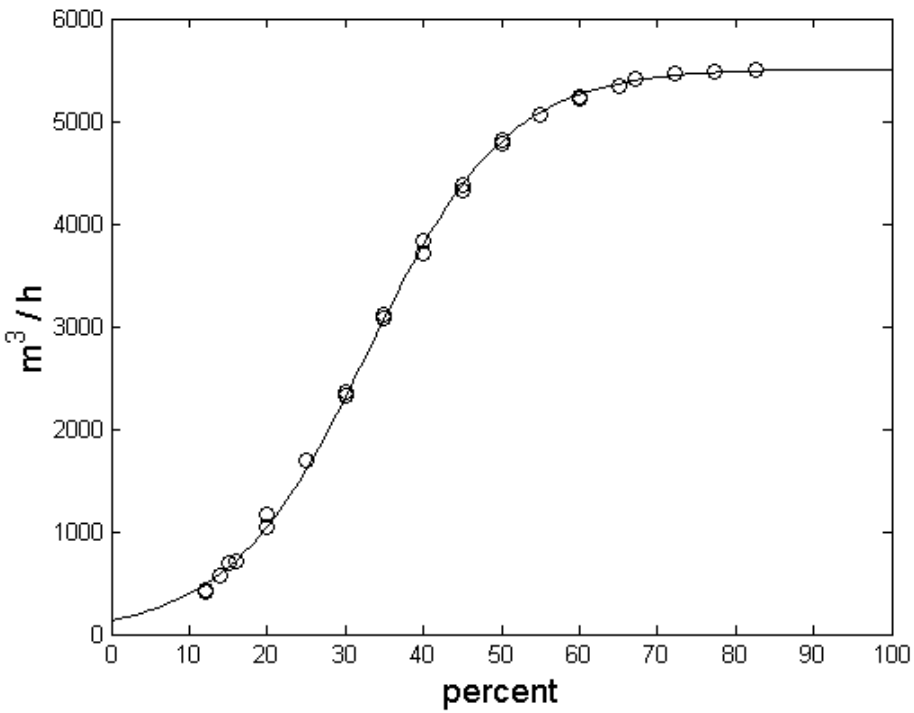
- The **system** is also commonly called the plant or the process
- The **desired output response** is also called the set point, the command input or the reference level – essentially, this is how we want the system to behave
- The **output** is the (measured) response of the system – this is how the system actually behaves!
- The **controller** attempts to adjust the **control input** to minimise the difference between the actual output and the desired output

Environmental Control

Environmental Control

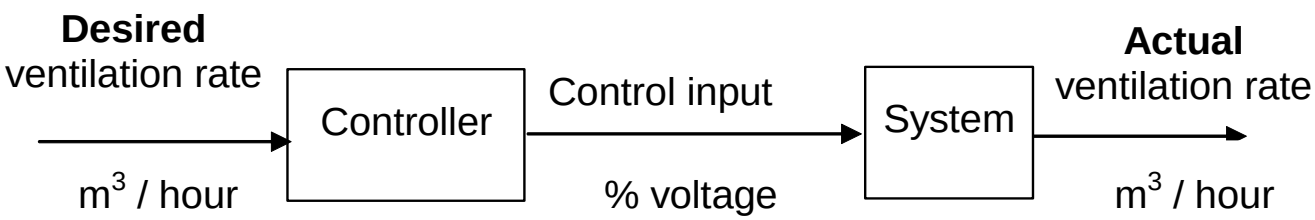
Ventilation Rate in a Building





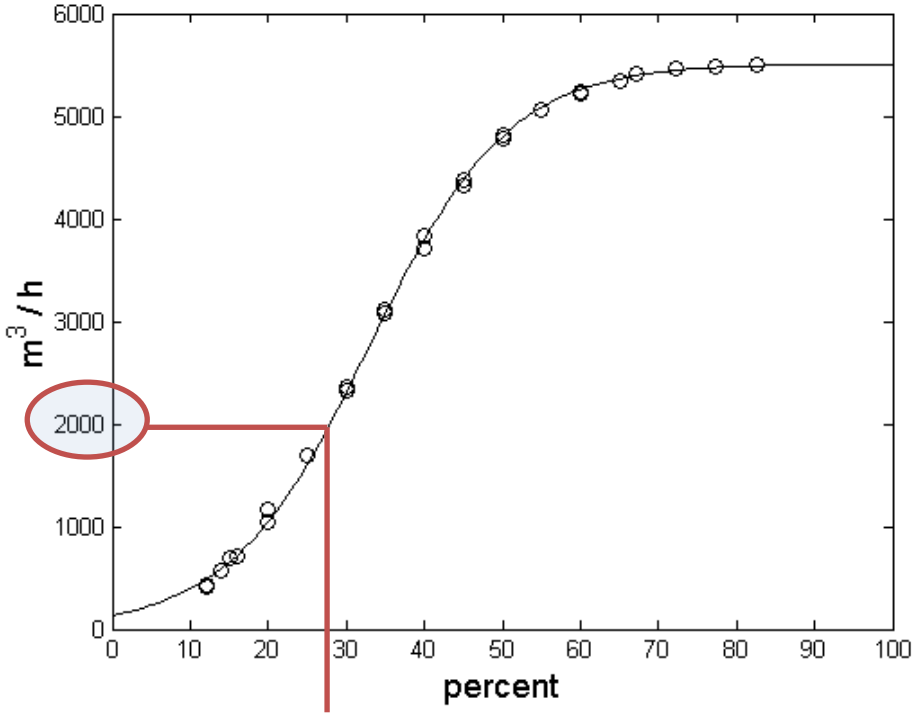
Steady state **power curve** shows the relationship between applied **voltage** to an axial fan and the measured **ventilation rate**

Open–Loop control of ventilation rate



- An open–loop controller is obtained from the steady state power curve
- Example – if the desired ventilation rate is chosen as 2000 m^3 / hour then the control input should be 28% voltage (check the graph)

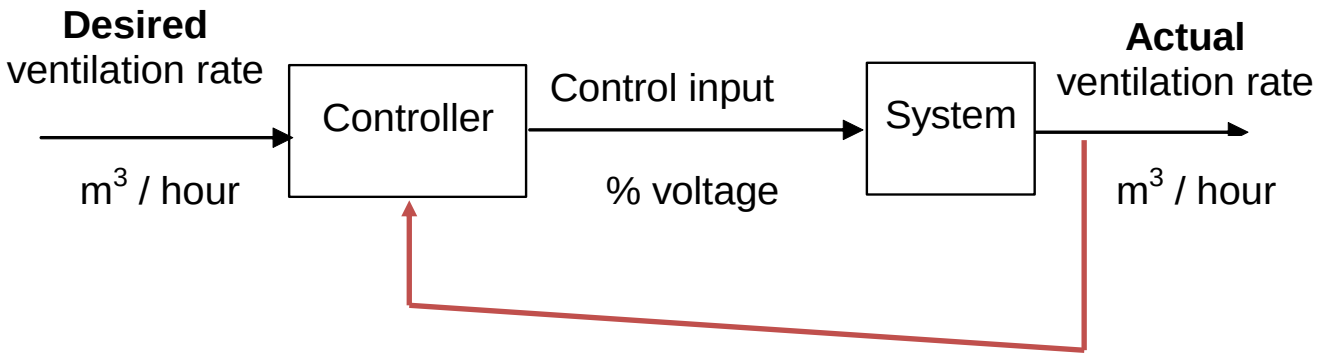
Open–loop control of ventilation rate



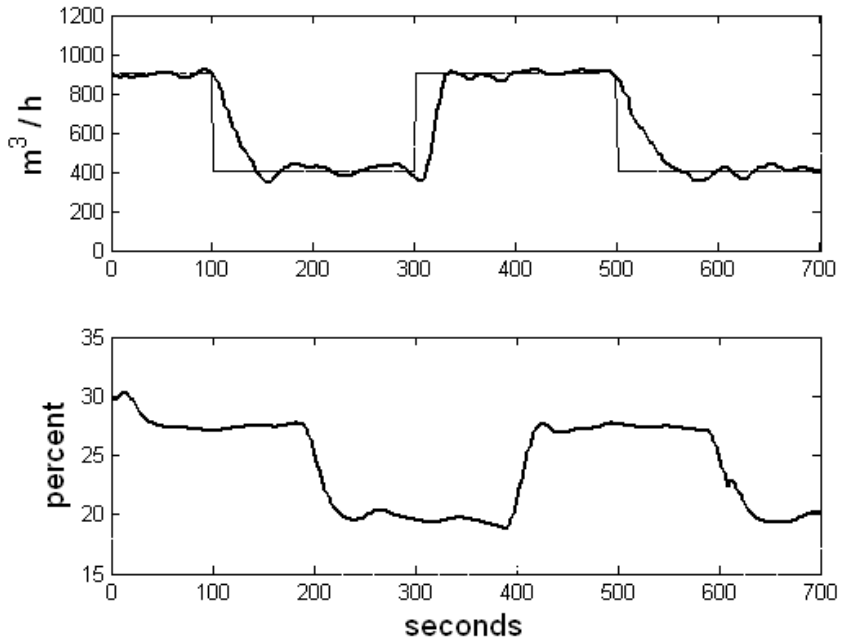
What are the limitations of this approach?

Advertisement from a company manufacturing fans for animal houses

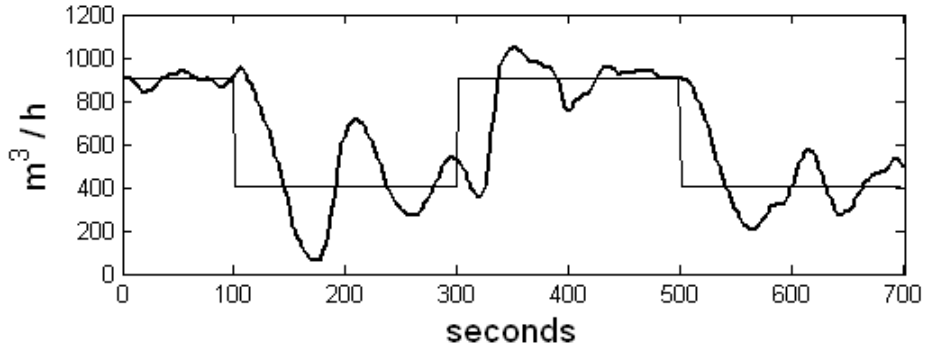
- "Enhance your yield and company profits!"
- "Your pigs need never lie in a draft again!"
- "Minimise heating costs"
- "Outside influences no longer effect the ventilation rate"
- "Fancom have introduced Exavent, a breakthrough in exact ventilation control”



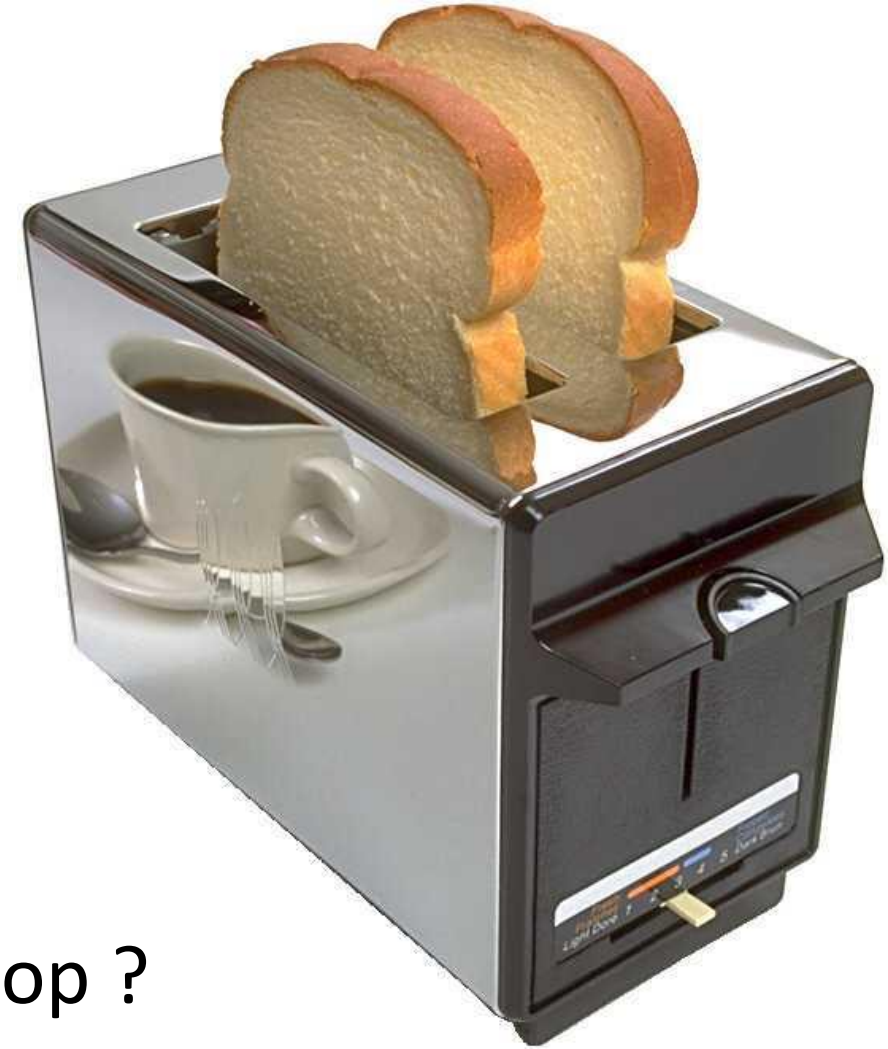
Closed–Loop Control of Ventilation Rate



- Results from a Year 3 student research project
- Initial design shown below, improved controller left



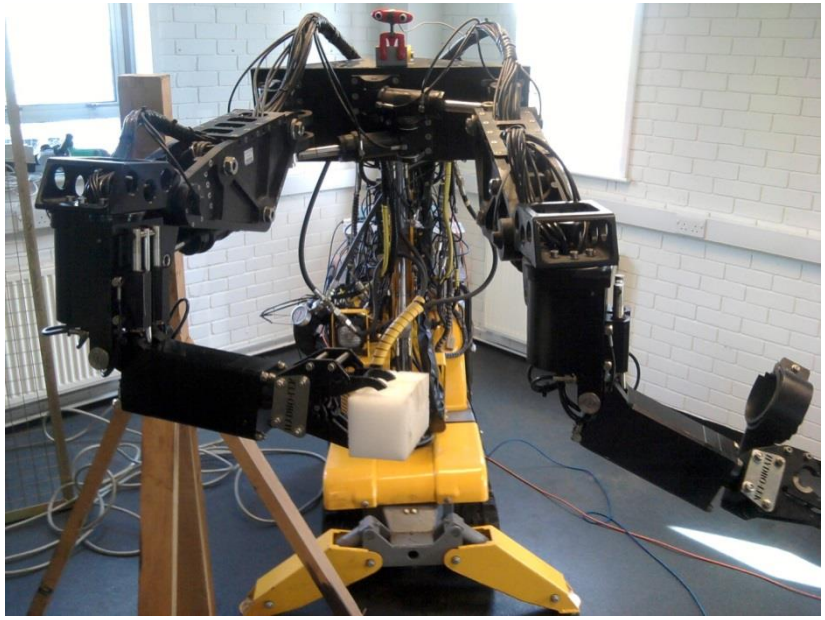
STOP
REHEAT
FROZEN
1
2
3
4
5
6



Open or Closed–Loop ?

Exercise

Research into Mobile Robots for Unstructured Environments



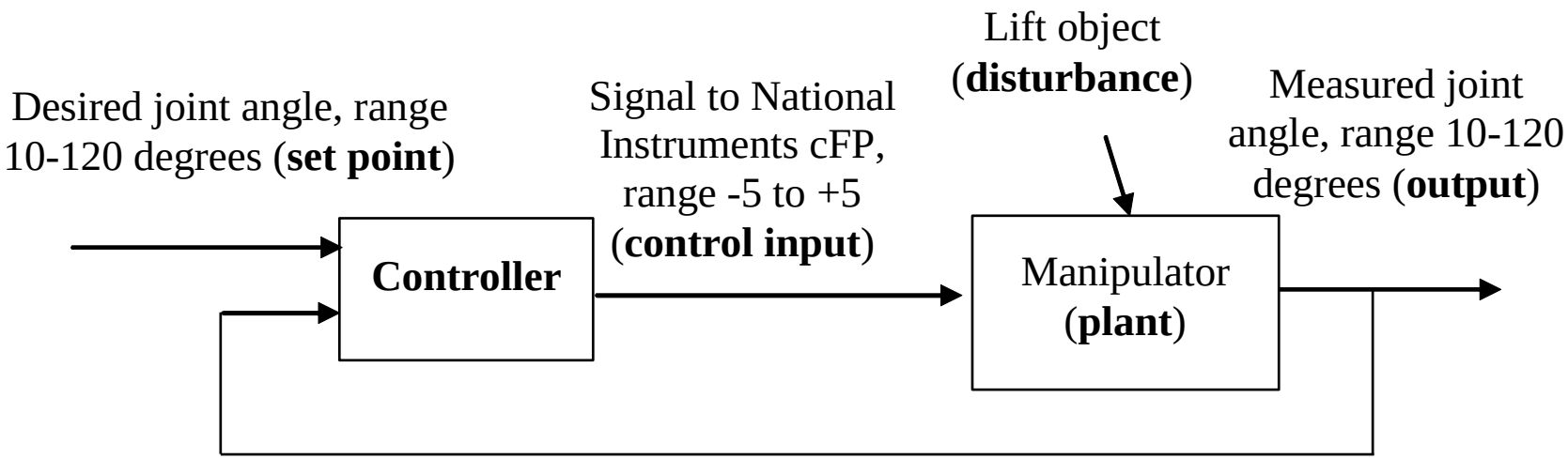
Nuclear Decommissioning Authority

NATIONAL NUCLEAR LABORATORY

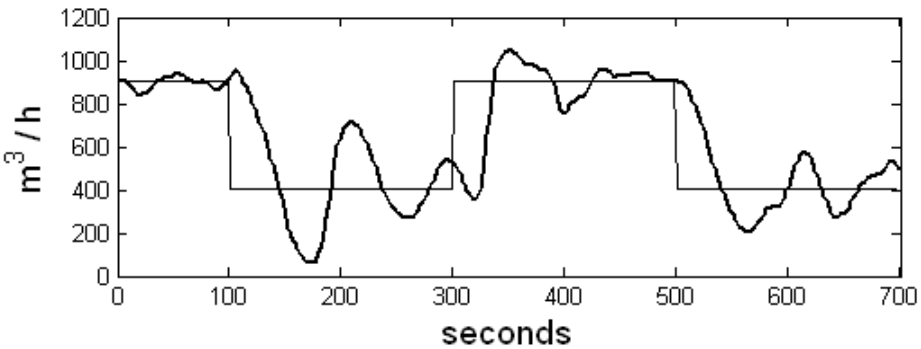
Exercise

- A robotic manipulator is hydraulically actuated. Consider movement of one joint in isolation. For R&D purposes, National Instruments (NI) software on a PC is used to control the robot. The PC transmits information to a NI Compact Fieldpoint Real-Time controller (cFP) linked to the hydraulics. For control design purposes on the PC, this information is expressed as a **signal in the range -5 to +5** where positive and negative implies the joint will "open" or "close" respectively (and zero means stop moving).
- A potentiometer feedback sensor measures the **joint angle**, returning a calibrated signal in the range **10 to 120 degrees**. An algorithm on the PC attempts to control this angle, i.e. so that it follows a pre-defined trajectory.
- The manipulator is used to first move an empty gripper (the end effector) and subsequently to first grasp and lift a heavy object.
- Your task: **draw a block diagram of the closed-loop control system** for regulating joint angle (consider just one joint), identifying the plant, controller, output, control input, disturbance input and the set point, and noting the practical range of any signals where appropriate.

Exercise

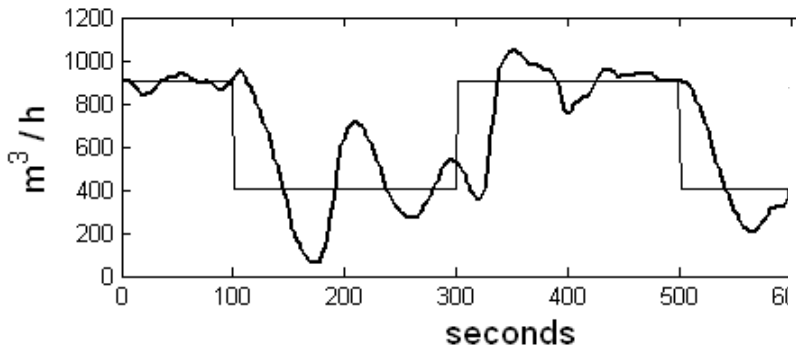

Desired joint angle, range 10-120 degrees (**set point**)
Controller
Signal to National Instruments cFP, range -5 to +5 (**control input**)
Lift object (**disturbance**)
Manipulator (**plant**)
Measured joint angle, range 10-120 degrees (**output**)

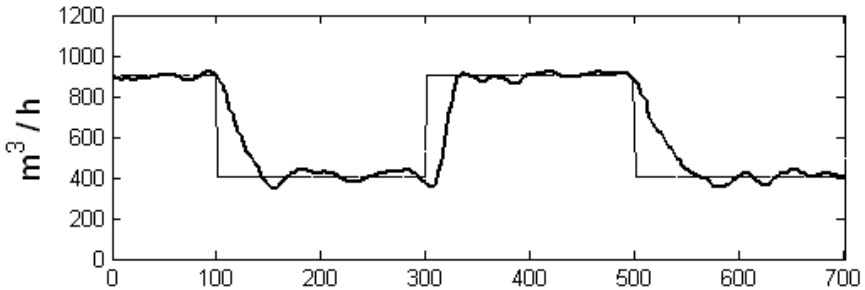
Control Objectives



Control Objectives

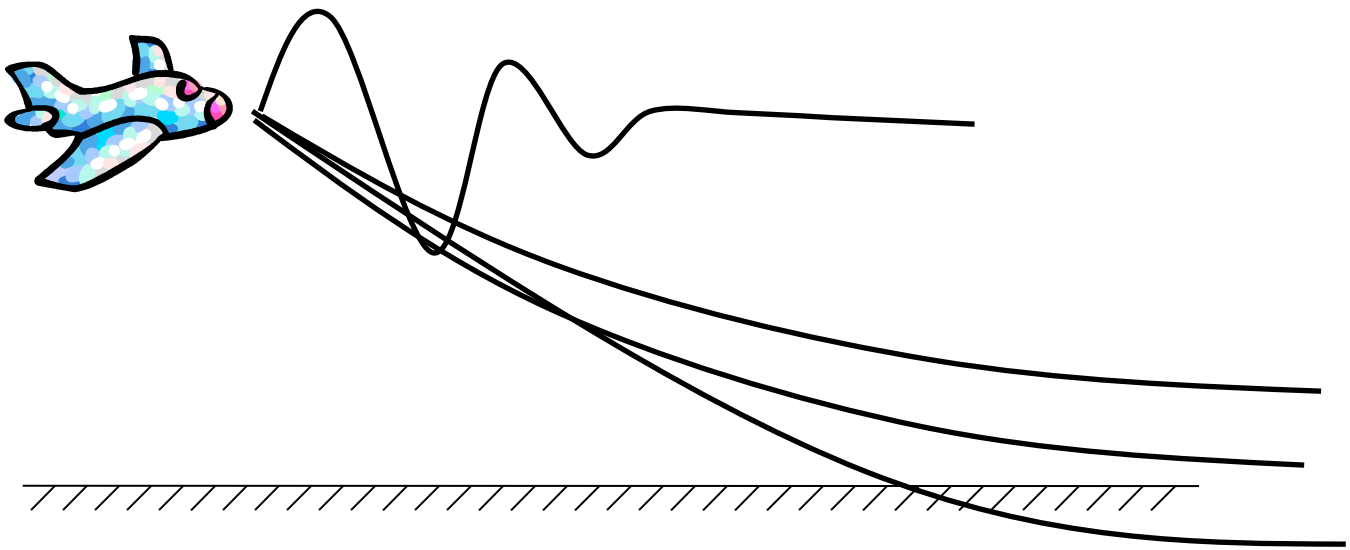
- The closed–loop system should be **stable**
- The output should track the desired output response in the **steady state**, i.e. there should be no steady state error
- The **transient behaviour** of the system should be satisfactory
- The control system should reject **disturbances**





Control Objectives

- How do these objectives relate to the altitude of a passenger aircraft on final approach to the runway in fog?



Control Objectives

- How do the flight control objectives of a military aircraft differ from a commercial passenger jet?

Advantages of Feedback

- Improved transient response
- Decreased sensitivity to modelling errors
- Decreased sensitivity to disturbances
- Ability to **eliminate steady state errors**
- Stabilisation of an unstable system

Disadvantages of Feedback

- Need to measure the output
- May be more difficult and expensive to design and construct
- Poorly designed feedback controllers can make things worse!

Next week's Lecture – First order systems and Embedded systems introduction (Arduino) to support your lab classes

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